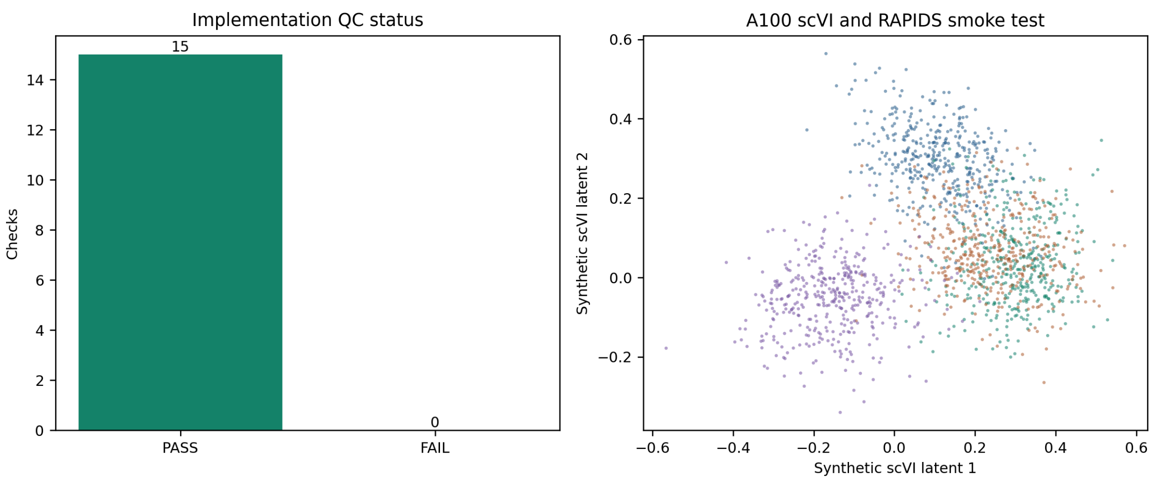
gdTAI V4.2 sidecar integration implementation QC

Decision

PASS_REVIEW_REQUIRED. The sidecar runner is implemented and passed deterministic unit, fail-closed, sparse-H5AD, direct-CUDA scVI, and RAPIDS clustering tests. No project-data integration, project-data scVI fit, pseudo-label construction, or classifier fit was performed.

- Checks passed: **15/15**.
- Synthetic scVI device: **cuda:0** on **NVIDIA A100 80GB PCIe**.
- Synthetic latent matrix: **1,600 x 8**, all finite.
- Locked cohorts remain excluded by code and the project-data stage aborts while its separate approval file is absent.
- CPU fallback is forbidden for scVI and RAPIDS stages.



Implementation QC

Frozen executable design

1. Select 4,000 common non-TCR, non-mitochondrial, non-ribosomal, and non-immunoglobulin HVGs from a deterministic source-balanced sample capped at 20,000 cells per source.
2. Stage only development cohorts in a new sparse H5AD on SSD. Locked cohorts are excluded before metadata loading, HVG selection, matrix staging, scVI, clustering, and label construction.
3. Fit the precommitted 30-dimensional negative-binomial scVI model on one A100 GPU, with no CPU fallback.
4. Run nine global and nine boundary RAPIDS Leiden partitions. The boundary is defined only by global clusters containing both independent primary-NK and productive-TCR anchors.
5. Select low-weight pseudo-NK development candidates only through repeated cluster agreement, independent-anchor purity, productive-TCR contamination, and cross-source representation. No CD3, TRDC, TRDV, NKG7, GNLY, KLRD1, TYROBP, FCER1G, or FCGR3A threshold defines truth.
6. Source balance and the primary-NK effective-weight cap are enforced at the later classifier-fitting gate. Pseudo-labels cannot control validation guardrails.

The two implementation safety floors frozen before project-data execution are at least 20 independent anchors and at least 10 primary NK anchors per qualifying cluster. They prevent tiny-anchor clusters from passing a percentage rule by chance.

Checks

check	status	detail
checksum_bound_preflight_approval	PASS	2026-08-08T08:29:03+08:00
development_locked_role_split	PASS	6 development objects; 3 locked objects
locked_development_permissions	PASS	locked cohorts have zero development permissions
cpu_fallback_forbidden	PASS	scVI and RAPIDS project stages require CUDA
marker_threshold_truth_forbidden	PASS	cluster evidence only
pseudo_labels_cannot_control_guardrails	PASS	primary truth retains guardrail ownership
python_compile	PASS	three modules compiled
deterministic_unit_and_sparse_h5ad_tests	PASS	11 passed in 2.52s
runner_validate_stage	PASS	checksum and role validation passed
project_data_stage_fails_without_approval	PASS	prepare aborted before SSD access
all_source_h5ads_unchanged	PASS	9/9 size/mtime pairs unchanged
direct_cuda_a100	PASS	NVIDIA A100 80GB PCIe; 79.3 GiB
synthetic_scvi_gpu_fit	PASS	latent=[1600, 8]; device=cuda:0
synthetic_rapids_neighbors_leiden	PASS	clusters=4,4
synthetic_gpu_no_cpu_fallback	PASS	direct CUDA only

Supervision gate

Project-data integration is still blocked. Activating `PROJECT_DATA_INTEGRATION_APPROVAL.json` authorizes only development-data sparse staging, scVI, consensus clustering, and pseudo-label QC. It does not authorize classifier fitting, threshold selection, model promotion, release fitting, or whole-atlas inference.

Exact artifacts

- Execution config: `configs/models/gdtai/v4_2_integration_execution.json`
- Runner: `workflows/gdtai/run_gdtai_v4_2_nk_reference_integration.py`
- Core: `workflows/gdtai/gdtai_v4_2_integration_core.py`
- Approval template: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_implementation_qc/PROJECT_DATA_INTEGRATION_APPROVAL_TEMPLATE.json`
- Full checks: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_2_implementation_qc/implementation_qc_checks.csv`